

1. Spam filtering for e-mails can be seen as an example of which of the following types of approaches of data mining?

- A) reduction B) association C) cause-and-effect modeling **D) classification**

2. _____ is a collection of techniques that seek to group or segment a collection of objects or observations into subsets, such that those within each subset are more closely related to one another than objects assigned to different subsets.

- A) Association rule mining B) Discriminant analysis
C) Cluster analysis D) Logistic regression

3. Which of the following is the first stage of joining clusters in agglomerative hierarchical clustering?

- A) by separating clusters into finer groups
B) by joining two clusters farthest away from each other
C) by joining two clusters that are not at a Euclidean distance
D) by joining two clusters that are closest to each other

4. In classification, which of the following would be considered as a categorical variable of interest for a credit approval decision for a requester?

- A) age of the requester
B) income of the requester revolving balance of the requester C) **reject or accept credit approval**

5. Which of the following data sets provides the most realistic estimate of the performance of a model on completely unseen data?

- A) validation data set **B) test data set**
C) training data set D) linear regression data set

6. _____ is the ratio of the number of transactions that include all items in the consequent as well as the antecedent to the number of transactions that include all items in the antecedent. A) Lift

- B) Logit
C) Support for the association rule
D) Confidence of the association rule

7. Divisive clustering method is different from agglomerative clustering methods in that divisive clustering methods _____.

- A) can only have a pair of subjects in each cluster
B) separate objects into a particular cluster in one step
C) separate n objects successively into finer groupings
D) can only have a single subject in each cluster

8. U-botit.com is an electronic commerce company that sells music online. It keeps a tab of what genre of music their registered customers buy. If u-botit.com were to use the data mining approach of association, which of the following actions would it take?

- A) send recommendations to customers based on their buying habits**
- B) conduct surveys to customers to gauge customer satisfaction**
- C) classify the customers based on the genre of music they listen to**
- D) provide discounts to certain genre buyers where sales are less**

9. Which of the following is a classification machine learning problem?

- A) predicting weight of mangoes**
- B) Predicting mango breed**
- C) Predicting number of leaves on a mango tree**
- D) Predicting number of days a mango will take to ripen**

10. Identify the following as a particular type of machine learning task? Given some description of a house, predict its cost

- A) Regression**
- B) Classification**
- C) Unsupervised Learning - Clustering**
- D) Not a machine learning task**

11. Identify the following as a particular type of machine learning task ?

- Fraud detection: Identify credit card transactions (for instance) which may be fraudulent in nature
- A) Regression**
 - B) Classification**
 - C) Unsupervised Learning - Clustering**
 - D) Not a machine learning task**

12. Identify the following as a particular type of machine learning task ?

You have bunch of photos of 6 people but without information who is on which one and want to divide this dataset into 6 piles, each with photos of one individual

- A) Regression**
- B) Classification**
- C) Unsupervised Learning - Clustering**
- D) Not a machine learning task**

13. Identify the following as a particular type of machine learning task ?

You have molecules, part of them are drugs and part are not but you do not know which are which and you want algorithm to discover the drugs

- A) Regression**
- B) Classification**
- C) Unsupervised Learning - Clustering**
- D) Not a machine learning task**

14. Identify the following as a particular type of machine learning task ?

Medical diagnosis: Diagnose a patient as a sufferer or non-sufferer of some disease

- A) Regression**
- B) Classification**
- C) Unsupervised Learning - Clustering**
- D) Not a machine learning task**

15. Identify the following as a particular type of machine learning task ? Given some description of a person, predict their income

A) Regression

B) Classification

C) Unsupervised Learning - Clustering

D) Not a machine learning task

16. Identify the following as a particular type of machine learning task ?

Given some description of a song, predict the year it was released

A) Regression

B) Classification

C) Unsupervised Learning - Clustering

D) Not a machine learning task

17. Identify the following as a particular type of machine learning task ?

Optical character recognition: Categorize images of handwritten characters by the letters represented A)

Regression

B) Classification

C) Unsupervised Learning - Clustering

D) Not a machine learning task

18. Identify the following as a particular type of machine learning task ?

A friend invites you to his party where you meet totally strangers. Now you will classify them (no prior knowledge) and this classification can be on the basis of gender, age group, dressing, educational qualification or whatever way you would like.

A) Regression

B) Classification

C) Unsupervised Learning - Clustering

D) Not a machine learning task

19. Identify the following as a particular type of machine learning task ?

NASA discovers new heavenly bodies and finds them different from previously known astronomical objects - stars, planets, asteroids, blackholes etc. (i.e. it has no knowledge about these new bodies) and classifies them the way it would like to (distance from Milky way, intensity, gravitational force, red/blue shift or whatever)

A) Regression

B) Classification

C) Unsupervised Learning - Clustering

D) Not a machine learning task

20. Identify the following as a particular type of machine learning task ? Face

detection: Find faces in images (or indicate if a face is present)

A) Regression

B) Classification

C) Unsupervised Learning - Clustering

D) Not a machine learning task

21. Identify the following as a particular type of machine learning task ?

Estimate the relationship between the stock market index and it's relationship with the macro economic variables.

A) Regression

B) Classification

C) Unsupervised Learning - Clustering

D) Not a machine learning task

22. Identify the following as a particular type of machine learning task ?

Examining the exchange rate movement and its dependency on several key macro economic factors .

A) Regression

B) Classification

C) Unsupervised Learning - Clustering

D) Not a machine learning task

23. Identify the following as a particular type of machine learning task ? Spam

filtering: Identify email messages as spam or non-spam

A) Regression

B) Classification

C) Unsupervised Learning - Clustering

D) Not a machine learning task

24. Identify the following as a particular type of machine learning task ?

Let's suppose you have never seen a Cricket match before and by chance watch a video on internet, now you can classify players on the basis of different criterion: Players wearing same sort of kits are in one class, Players of one style are in one class (batsmen, bowler, fielders), or on the basis of playing hand (RH vs LH) or whatever way you would observe [and classify] it.

A) Regression

B) Classification

C) Unsupervised Learning - Clustering

D) Not a machine learning task

25. Identify the following as a particular type of machine learning task ?

We are conducting a survey of 500 questions about predicting the IQ level of students in a college. Since this questionnaire is too big, so after 100 students, administration decides to trim the questionnaire down to fewer questions and for it we use some statistical procedure trim it down.

A) Regression

B) Classification

C) Unsupervised Learning - Clustering

D) Not a machine learning task

26. Identify the following as a particular type of machine learning task ?

Customer segmentation: Predict, for instance, which customers will respond to a particular promotion A)

Regression

B) Classification

C) Unsupervised Learning - Clustering

D) Not a machine learning task

27. Identify the following as a particular type of machine learning task ?

Weather prediction: Predict, for instance, whether or not it will rain tomorrow A)

Regression

B) Classification

C) Unsupervised Learning - Clustering

D) Not a machine learning task

28. How does the ensemble work?

a) It uses several machine learning models with different algorithms to improve the accuracy of prediction

b) It uses Several machine learning models with identical algorithms to improve the accuracy of prediction

A) both statement a & b are false

B) statement a alone is true C)

both statement a & b are true

D) statement b alone is true

29. We have a training data is of size N. Draw random sample with replacement of size N – This gives a new dataset, it might have repeated observations, We are selecting records one-at-a-time, returning each selected record back in the population, giving it a chance to be selected again. what is this methodology?

A) Bagging

B) Boosting

30. Choose all those which are applicable for Boosting:

A) Time and computationally expensive

B) Easy to implement in all scenarios

C) Complexity of the classification increases

A) statement a and statement b

B) statement b

C) statement a and c

D) statement a

31. Where do the multiple layers of machine learning models are placed one over another where each of the models passes their predictions to the model in the layer above it and the top layer model takes decisions based on the outputs of the models in layers below it.

A) Bagging

B) Boosting

C) Stacking

D) Averaging

32. Ensembling helps to increase the interpretation of the business problem easily:

A) True

B) False

C) Not applicable

33. Statement 1. Combining multiple models of same ML algorithms gives better predictions. Statement 2. Combining multiple predictions generated by different algorithms would give better predictions. **A) Both the statements are True but depends on the type of Problem as well.**

B) Statement 1 is False and statement 2 is true

C) Statement 2 is False and statement 1 is true

D) Statement 2 alone is True

34. Bagging helps in reducing the bias in the model

A) True

B) False

35. In Boosting, we use sequential learning to correct previous models weakness?

A) True

B) False

36. Bagging of an unstable algorithm helps to stabilise and generalise the model?

A) True

B) False

37. Ensembles is not recommended to use when there is significant diversity among the models?

A) True

B) False

38. Sentiment Analysis is an example of:

I. Regression

II. Classification

~~III. Clustering~~

IV. Reinforcement Learning

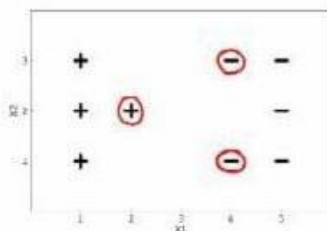
a) only I

b) I and II

b) II and III

d) I, II and III ✓

39. Suppose you are using a Linear SVM classifier with 2 class classification problem. Now you have been given the following data in which some points are circle red that are representing support vectors.



If you remove the non-red circled points from the data, the decision boundary will change?

- a) **True** b) False c) None d) Can't say

40. On what factor the number of outputs in neural network depends?

- a) distinct inputs **b) distinct classes** c) both on distinct classes & inputs d) none of the mentioned

41. In perception learning, what happens when input vector is correct

- a) small adjustment in weight is done b) large adjustments in weight is done
c) **no adjustments in weight is done** d) weight adjustments doesn't on classification of input vector

42. What kind of learning is involved in pattern clustering task?

- a) supervised **b) unsupervised** c) learning with critic d) none of the mentioned

43. What is the feature of ANNs due to which they can deal with noisy, fuzzy, inconsistent data?

- a) associated nature of networks b) distributive nature of networks
b) **both associative & distributive** d) none of the mentioned

44. A feature F1 can take certain value: A, B, S, D, E & F and represents grade of students From a college.

Which of the following statements is true in following case? a)

Feature F1 is an example of nominal variable.

- b) Feature F1 is an example of ordinal variable.**
c) It doesn't belong to any of the above category.
d) Both of these.

45. Imagine, you are working as Data Scientist and you want to develop a machine learning algorithm which predicts the number of views on the articles.

Your analysis is based on features like author name, number of articles written by the same Author in past and a few other features. Which of the following evaluation metric would you Choose in that case?

- a) Mean Square Error** b) Accuracy c) F1 Score d) All of the above

46. Imagine you are working on a project which is a binary classification problem. You trained a model on training dataset and get the below confusion matrix on validation dataset.

	Predicted:	
	NO	YES
Actual: NO	50	10
Actual: YES	5	100

Based on the above confusion matrix, choose the correct value for accuracy of the model? a) **approx.**

- 0.91** b) approx.0.09 c) approx.-0.85 d) approx.0.5

47. The process of adjusting the weight is known as?

- a) Activation b) Synchronisation c) **learning** d) none of the mentioned

48. Which of the following can be true for selecting base learners for an ensemble?

- I. Different learners can come from same algorithm with different hyper parameters
- II. Different learners can come from different algorithms
- III. Different learners can come from different training spaces

- a) 1 b) 2 c) 1 and 3 d) **1,2 and 3**

49. Which of the following algorithm doesn't use learning Rate as one of its hyper parameter? I.

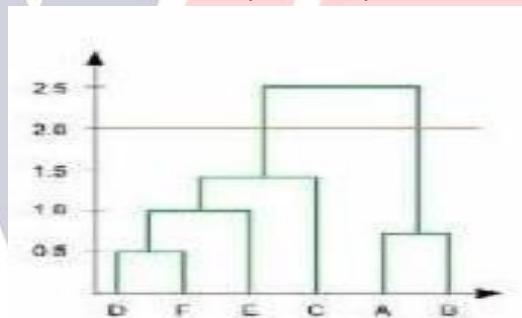
- Gradient Boosting II. Extra Trees III. AdaBoost IV. Random Forest

- a) 1 and 3 b) 1 and 4 c) 2 and 3 d) **2 and 4**

50. In random forest or gradient boosting algorithms, features can be of any type, For example, it can be a continuous features or a categorical feature. Which of the following option is true when you consider these types of features?

- a) Only Random forest algorithm handles real valued attributes by discretizing them
- b) Only Gradient boosting algorithm handles real valued attributes by discretizing them
- c) **Both algorithms can handle real valued attributes by discretizing them**
- d) None of these

51. In the figure below, if you draw a horizontal line on y-axis for $y=2$. What will be the number of clusters formed?



- a) 1 b) **2** c) 3 d) 4

52. Which of the following statement is true about k-NN algorithm?

- I. k-NN performs much better if all of the data have the same scale
- II. k-NN works well with a small number of input variables (p), but struggles when the number of inputs is very large
- III. k-NN makes no assumptions about the functional form of the problem being solved

- a) 1 and 2 b) 1 and 3 c) Only 1 d) **All of the above**

53. Which of the following is true about Manhattan distance?

- a) **It can be used for continuous variables**
- b) It can be used for categories variables
- c) It can be used for categorical as well as continuous
- d) None of these

54. Which of the following algorithm is an example of ensemble learning algorithm?

- a) AdaBoost b) **Decision Tree** c) Knn classifier d) Support vector machine

55. Which of the following will be true about k in k-NN terms of Bias?

- a) When you increase the k the bias will be increase
- b) When you decrease the k the bias will be increase
- c) Can't say
- d) None of these

56. Fuzzy clustering creates

- a) hard clusters b) soft clusters c) hierarchical clusters d) None of the above

57. How the decision tree reaches its decision?

- a) Single test b) Two test c) Sequence of test d) No test

58. In which of the following learning the teacher returns reward and punishment to learner?

- a) Active learning b) Reinforcement learning
- c) Supervised learning d) Unsupervised learning

59. Support vectors are the data points that lie closest to the decision surface.

- A. True B. False C. None D. Can't say

60. Which of the following algorithm is most sensitive to outliers?

- A. K-means clustering B. K-medians clustering C. K-modes clustering D. K-medoids clustering

70. A 4-input neuron has weights 1,2,3 and 4. The transfer function is linear with the constant of Proportionality being equal to 2. The inputs are 4, 10, 5 and 20 respectively. The output will be:

- A, 238 B. 76 C. 119 D.123

71. Which of the following is a valid production rule for the decision tree below?



- A. IF Business Appointment = No & Temp above 70 = No
THEN Decision = wear slacks
- B. IF Business Appointment = Yes & Temp above 70= Yes
THEN Decision = wear shorts
- C. IF Temp above 70 = No
THEN Decision = wear shorts

D. IF Business appointment = No & Temp above 70= No
THEN Decision = wear jeans

72. Which of the following is not a machine learning algorithm?

- A. SVG B. SVM C. Random forest D. Neural Networks

73. Inductive learning involves finding a

- A. Consistent Hypothesis B. Inconsistent Hypothesis
C. Regular Hypothesis D. Irregular Hypothesis

74. If the training accuracy is 100% for a machine learning algorithm, but on validation set its accuracy is 70%, what does it signifies about the model learned?

- A. Under fitting B. Nothing, the model is perfect
C. Overfitting D. None of the above options

75. What is /are true about kernel in SVM?

- I. Kernel function map low dimensional data to high dimensional space II.
It's a similarity function

- A. I B. II C. I and II D. None of these

76. Is it possible that Assignment of observations to clusters does not change between successive iterations in KMeans

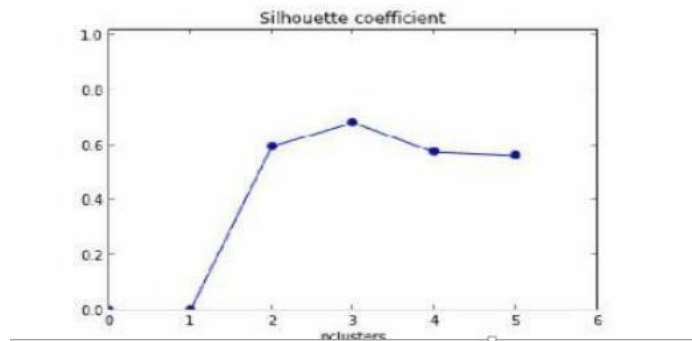
- A. Yes B. No C. Can't say D. None of these

77. Which of the following metrics, do we have for finding dissimilarity between two clusters in hierarchical clustering?

- I. Single-linkage II. Complete-Linkage III. Average – Linkage

- A. 1 and 2 B. 2 and 3 C. 1 and 3 D. 1,2 and 3

78. What should be the best choice of no. of clusters based on the following results:



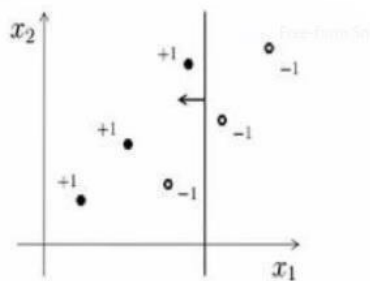
- A.1 B. 2 C. 3 D.4

79. Which of the following is true about training and testing error in such case?

Suppose you want to apply AdaBoost algorithm on Data D which has T observations. You set half the data for training and half for testing initially. Now you want to increase the number of data points for training $T_1, T_2, \dots, T_n$ where $T_1 < T_2 < \dots < T_{n-1} < T_n$.

- A. The difference between training error and test error increases as number of observations increases.
- B. The difference between training error and test error decreases as number of observations increases.
- C. The difference between training error and test error will not change.
- D. None of These

80. Consider the following figure for answering the next few questions. In the figure, X_1 and X_2 are the two features and the data point is represented by dots (-1 is negative class and +1 is a positive class). And you first split the data based on feature X_1 (say splitting point is x_{11}) which is shown in the figure using vertical line. Every value less than x_{11} will be predicted as positive class and greater than x_{11} will be predicted as negative class.



How many data points are misclassified in above image?

- A. 1
- B. 2
- C. 3
- D. 4

81. In which of the following scenario a gain ratio is preferred over information Gain?

- A. When a categorical variable has very large number of category
- B. When a categorical variable has very small number of category
- C. Number of categories is the not the reason
- D. None of these

82. In Decision Tree, by comparing the impurity across all possible splits in all possible Predictors, the next split is chosen. How we can measure the Impurity?

- A. UC
- B. Entropy, Gini-index
- C. ROC
- D. MAPE

83. Random Forest Modeling (Ensemble Modeling) uses

- A. Bagging
- B. Boosting
- C. Both of the above
- D. None

84. To identify the similar(neighbouring)records, K-Nearest Neighbours uses which distance metrics A. Euclidean distance B. Perpendicular distance C. Diagonal distance D. None

85. K-NN algorithm does more computation on test time rather than train time.

- A. True
- B. False
- C. Time is same for both testing and training
- D. None

86. The bagging is suitable for high variance low bias models?

I. True II. False

A. Only I

B. Only II

C. Can't say

D. Both I and II

